SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: **Database Management Systems**
COURSE CODE: **CSA05**

COURSE OUTCOME COVERED

CO1: *Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships.* (BL1, BL2)

Activity Tool : Concept Mapping (Medium)
Assessment Tool Weightage: 35%

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QUESTIONS		Total Marks: 50	
Q.No	**Question**	**Bloom's Level**	**Marks**
1	Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.	BL2 – Understand	**5**
2	Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.	BL2 – Understand	**5**
3	Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.	BL2 – Understand	**5**
4	Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints.	BL2 – Understand	**5**
5	Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.	BL2 – Understand	**5**
6	Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.	BL2 – Understand	**5**

7	Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval.	BL2 – Understand	**5**
8	Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.	BL2 – Understand	**5**
9	Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.	BL2 – Understand	**5**
10	Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.	BL1 – Remember	**5**

Code of Conduct:

I ___R harika (192572140)_________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

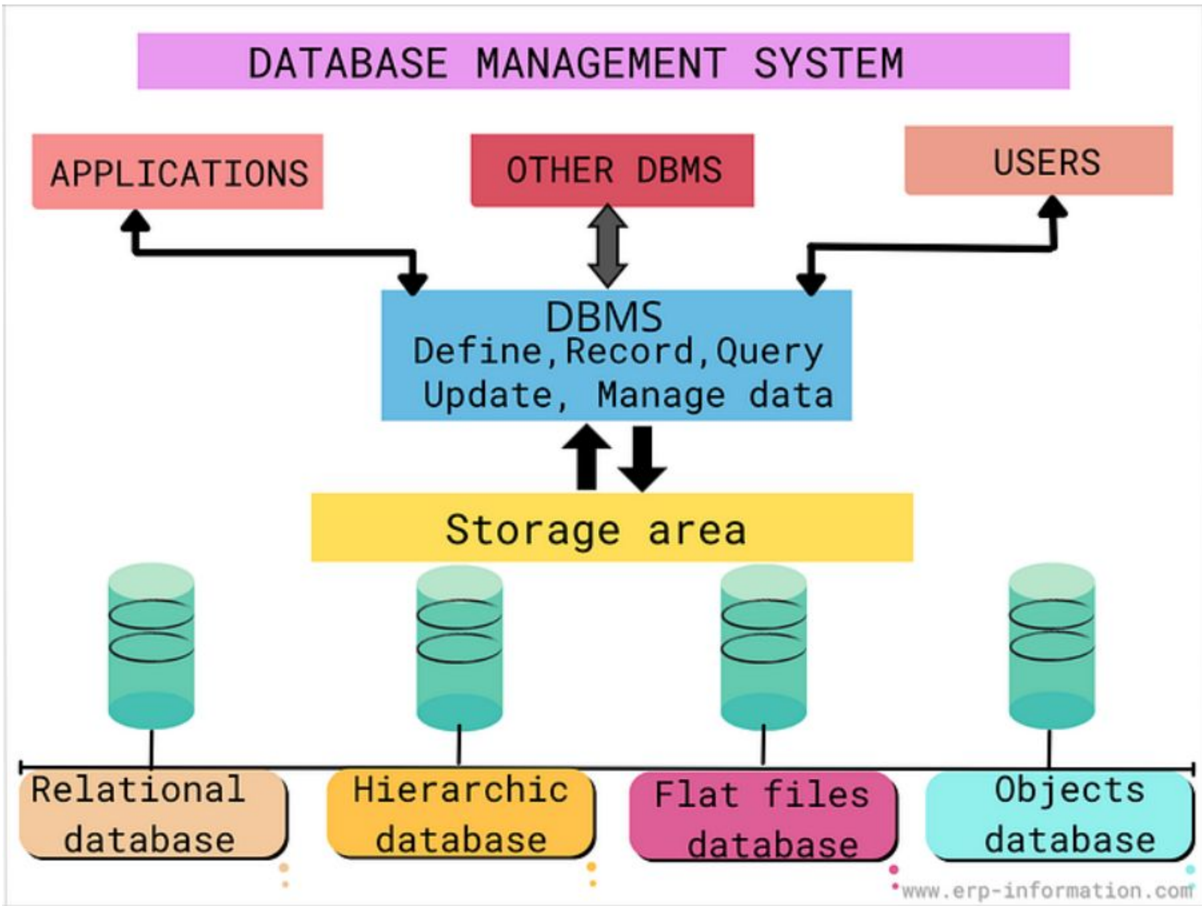
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

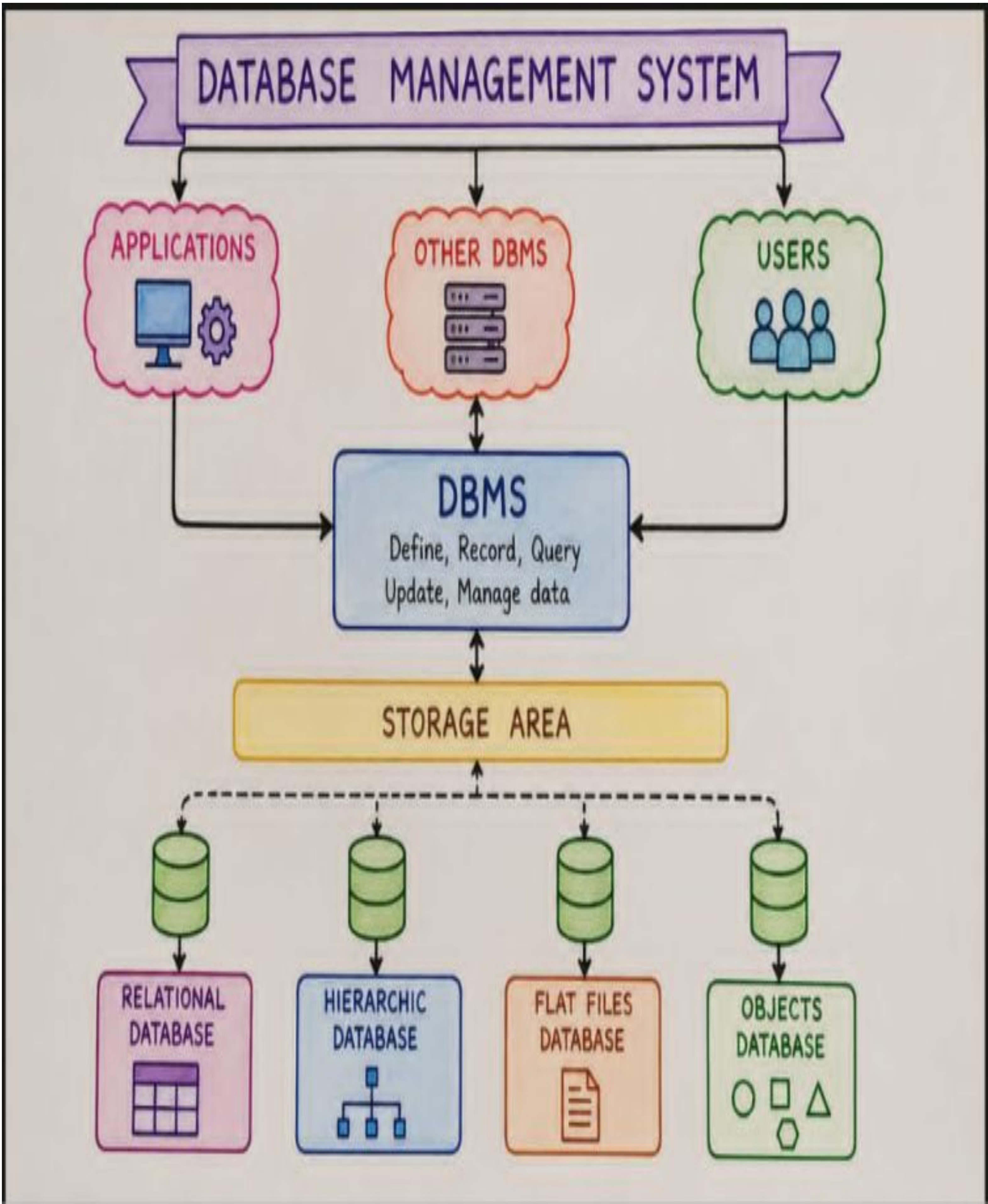
RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure

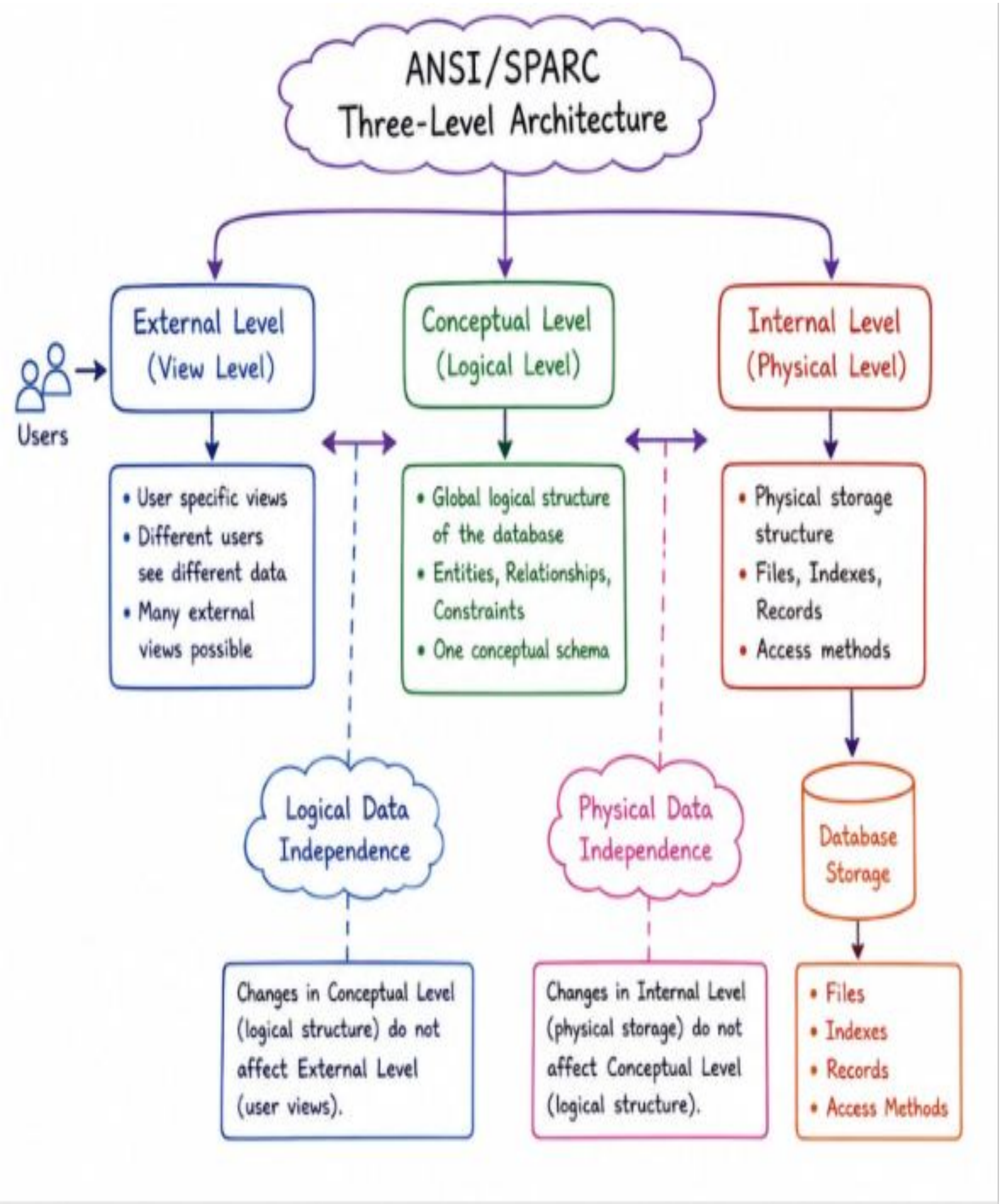
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand



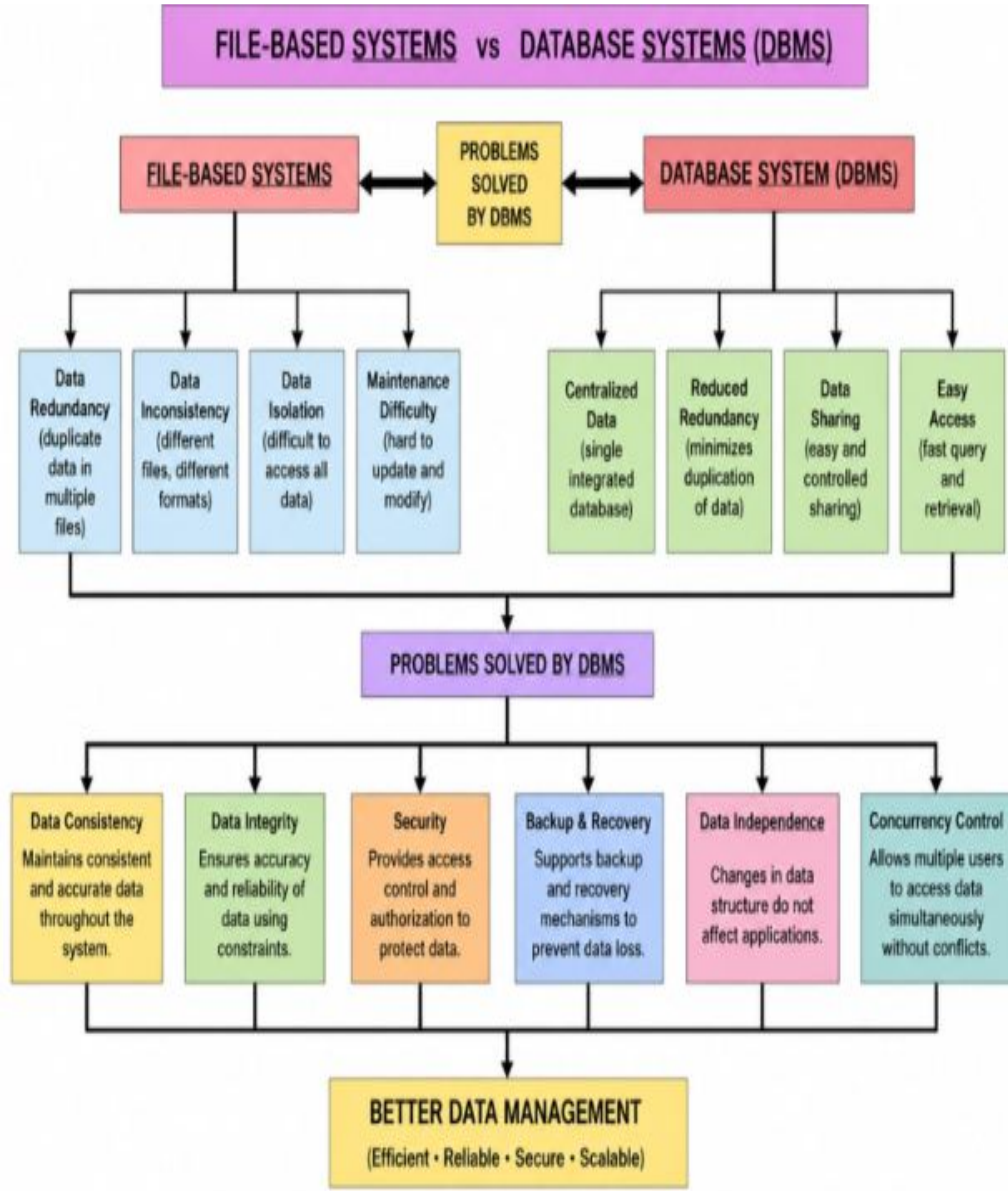
Concept map -1



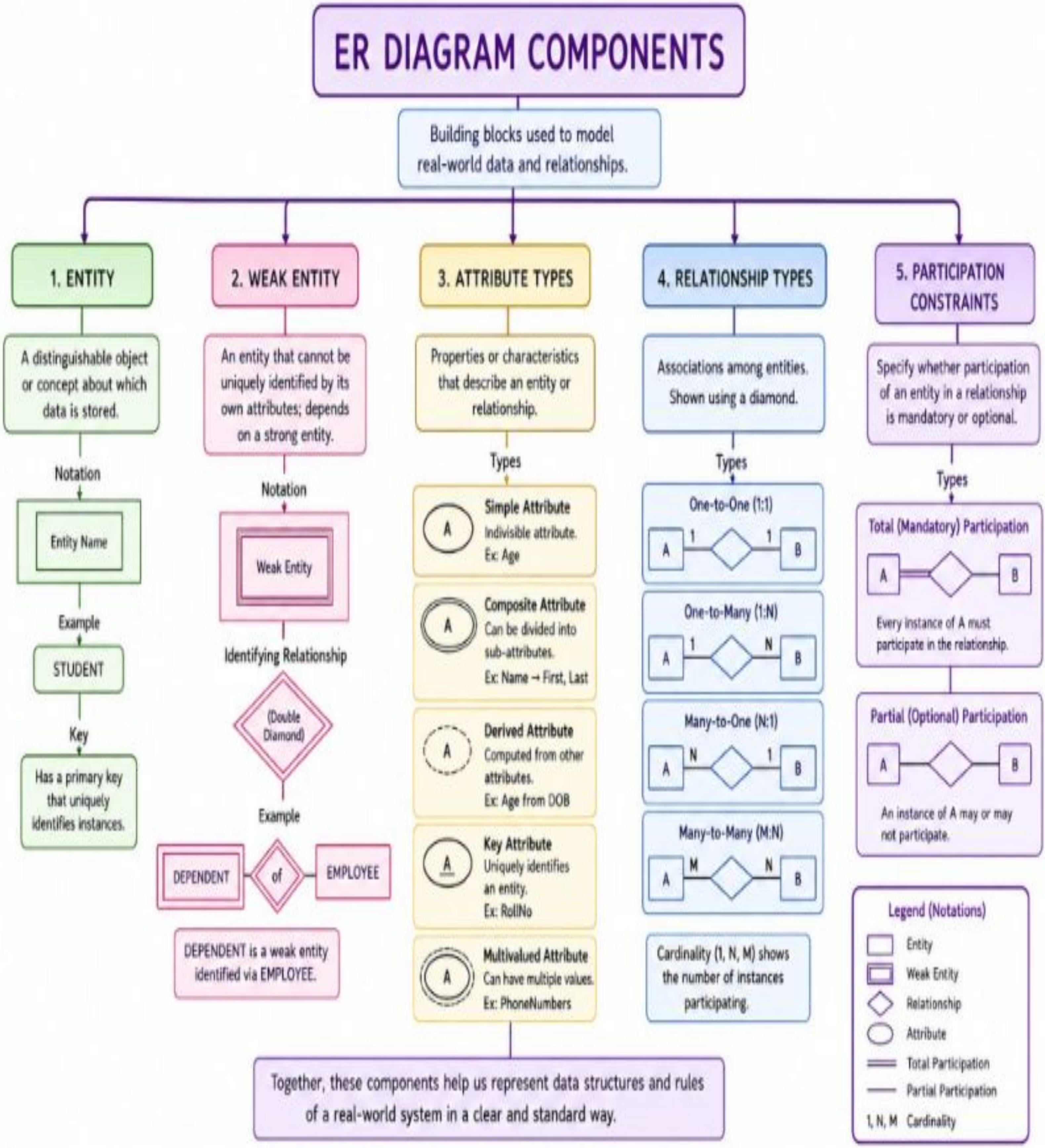
Concept map -2



Concept map-3



Concept map-4

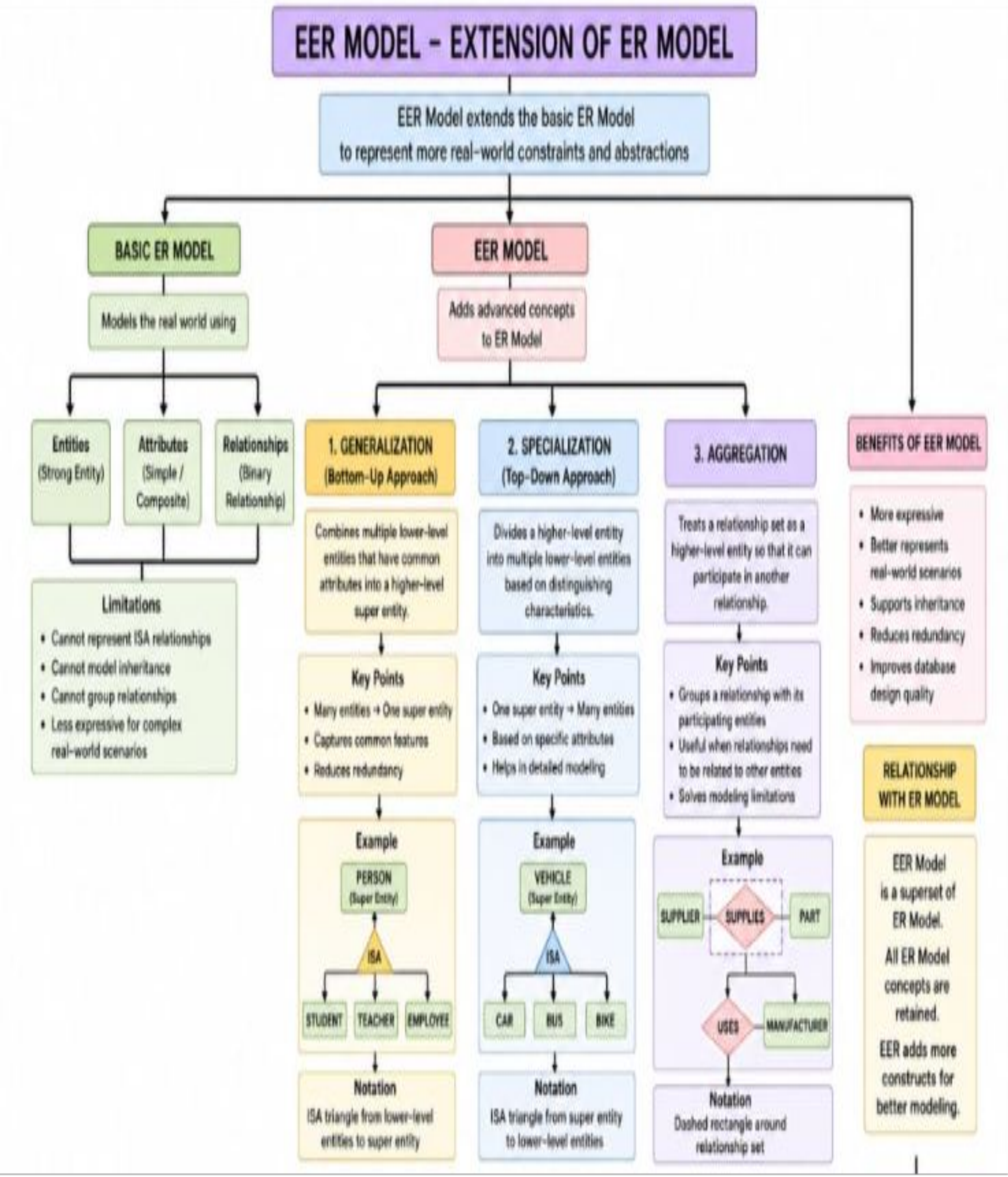


Concept map-5

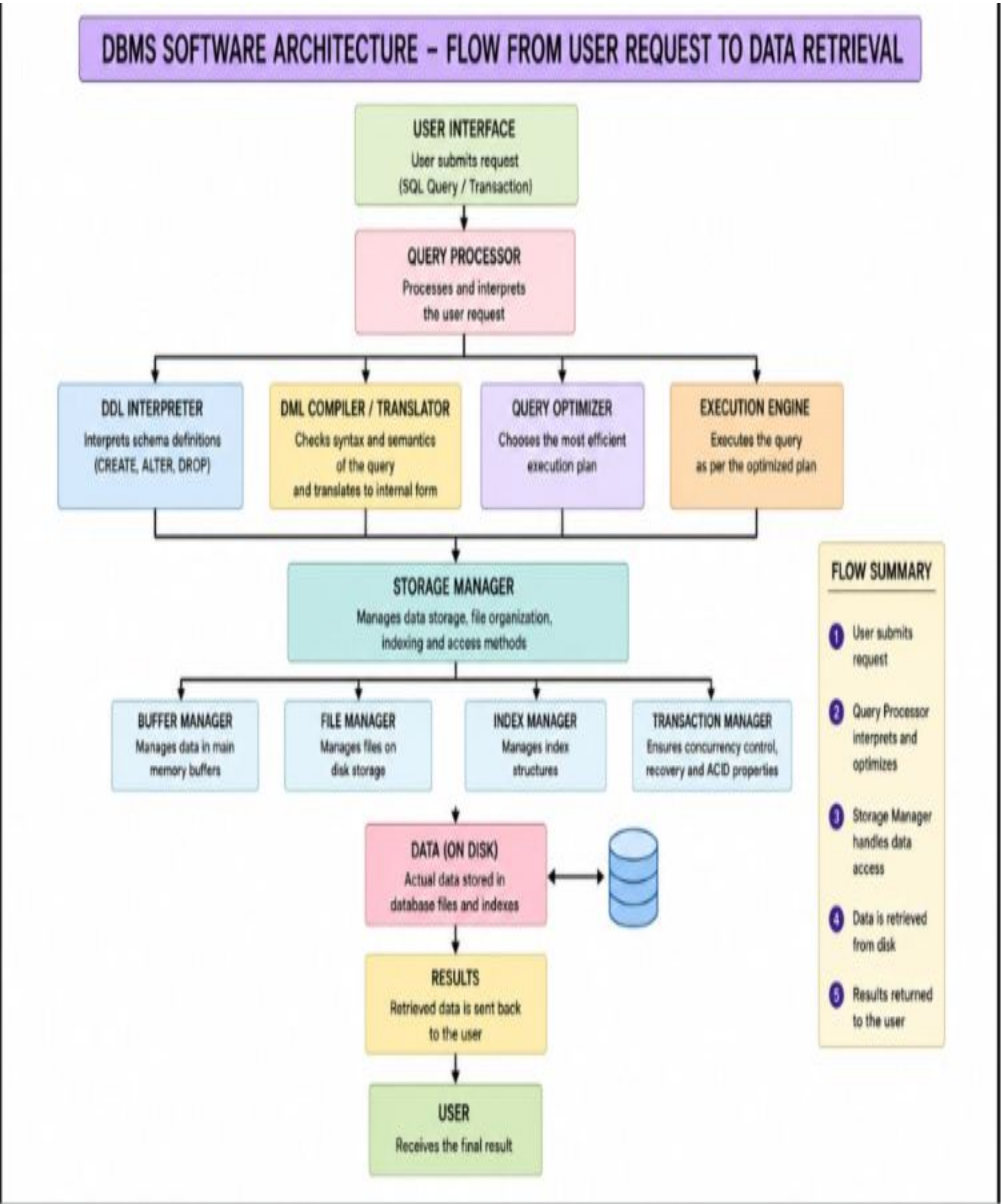
COMPARISON OF DATA MODELS

	HIERARCHICAL DATA MODEL	NETWORK DATA MODEL	RELATIONAL DATA MODEL
Structure	Organized in a tree structure. One root (parent) with multiple levels of child nodes.	Organized as a graph structure. Records can have multiple parents and multiple children.	Organized in tables (relations). Data stored in rows and columns with relationships using keys.
Key Properties	• Parent-child relationship • One-to-many relationship • Strict hierarchical structure • Each child has only one parent	• Many-to-many relationship • Records linked via sets • More flexible than hierarchical • Supports multiple parentage	• Data in tables (rows & columns) • Relationships using primary and foreign keys • Based on relational algebra • High data independence
Advantages	• Simple and easy to understand • Efficient for hierarchical data • Fast access to data	• More flexible than hierarchical • Handles complex relationships • Reduces data redundancy	• Easy to use and maintain • Data independence • Reduces redundancy • Powerful query capability (SQL)
Limitations	• Rigid structure • Difficult to handle many-to-many relationships • Changes in structure are hard	• Complex to design and navigate • Difficult to implement • Low data independence	• Performance overhead for large data • Complex joins may slow down queries
Example	File System Directory Structure	CODASYL Database System	MySQL, Oracle, PostgreSQL, SQL Server

Concept map -6



Concept map-7

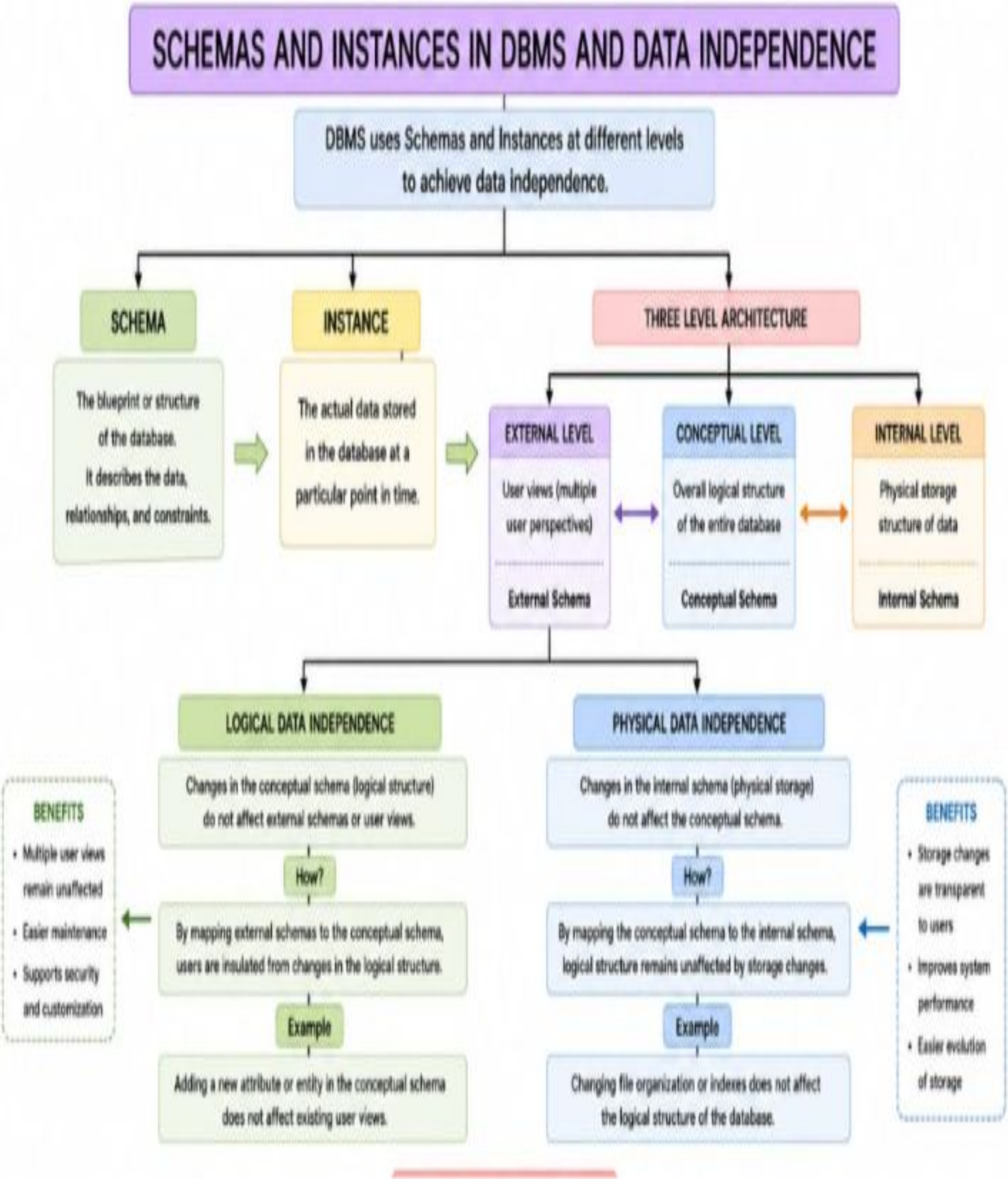


Concept map -8

DATA MODELS (CO1) MAPPED TO REAL-WORLD APPLICATIONS

DATA MODEL	REAL-WORLD APPLICATIONS (EXAMPLES)	WHY THIS MODEL IS SUITABLE
1. RELATIONAL DATA MODEL	Banking Systems Airline Reservation Systems E-commerce Platforms University Databases	• Structured data stored in tables (rows & columns) • Easy to query using SQL • Ensures data integrity using keys and constraints • Ideal for applications with structured data
2. HIERARCHICAL DATA MODEL	File Systems (OS directories) Organization Structure XML Data Representation Library Information Systems	• Data organized in a tree structure (parent-child) • Easy and fast access to data • Ideal for data with one-to-many relationships • Simple and efficient for hierarchical data
3. NETWORK DATA MODEL	Telecom Systems Airline Reservation (older systems) Supply Chain Management Hospital Information Systems	• Supports many-to-many relationships • More flexible than hierarchical model • Efficient for complex interconnected data • Reduces data redundancy
4. ENTITY-RELATIONSHIP (E-R) MODEL	Database Design (Conceptual Level) Business Process Modeling System Analysis & Design Data Warehouse Design	• Easy to understand and visualize data • Uses entities, attributes and relationships • Helps in designing databases before implementation • Platform independent (conceptual model)
5. OBJECT-ORIENTED DATA MODEL	CAD/CAM Systems Multimedia Applications Object-Oriented Programming Scientific Data Management	• Stores data as objects (with identity, methods) • Handles complex data (multimedia, images, 3D) • Supports inheritance & reusability • Best for modern applications and simulations

Concept map -9



Concept map-10